GoSpice architecture

A SPICE-compatible analog circuit simulator in pure Go, built as the fitness-evaluation engine for genetically evolved circuits

imports (never upward) · data flow · entry points · analysis / Kron · physics · GA layer · simulation core · leaf packages

`cmd/gospice`
CLI: simulate a netlist → CSV / SPICE3 raw
stdlib flag

`cmd/evolve`
CLI: run a GA experiment from JSON / YAML
writes runs/<timestamp>-<name>/

`examples/`
rc_lowpass · filter_topology · amplifier · solar
bifilar_coil · homopolar · arc_surge · deep_space …

evo/ — genetic algorithm layer
genome · encode_values (log scale, E12/E24/E96) · encode_topology (graph / Koza developmental) · operators · fitness · engine · checkpoint
Worker pool of runtime.NumCPU(), one simulator per worker · seeded *rand.Rand per generation and individual · results sorted by index
Failed circuits (singular, no convergence, floating node, timeout) → penalty fitness, never a crash · weighted sum, then NSGA-II

waveforms, TEMP, EMF (env) · lumped R/L/C/K (em)
→ wired into circuit.Builder by evo / cmd

env/ — outside influences, anywhere in the universe
Profile = composable Sources + circuit pose and motion
star (irradiance, wind, flares) · fields E(t), B(t) · thermal
radiation: TID dose, single events, neutrinos (computed, ≈ 0)
gravity: time dilation, redshift, Doppler, tidal limit
couple → device inputs + attributed ledger inflows · limits
presets: earth_noon · leo_storm · mars · europa · near_black_hole

analysis/ — analyses on the MNA system
op · dc sweep · ac (complex128 per frequency) · tran (adaptive step, LTE, breakpoints)
integrate: backward Euler · trapezoidal (default) · Gear-2
measure: rise time, gain, −3 dB bandwidth, overshoot, THD, power
power: quaternion + geometric-algebra power theory (→ hyper)
coupled: circuit ↔ plasma channel ↔ bubble ↔ light (→ plasma, fluid)

kron/ — Kron tensors & diakoptics
primitive Z / Y · connection C · $Z' = C^T Z C$
mesh & node forms, twin-tested vs. MNA (1e-9)
orthogonal · exterior open paths · TAN chords
diakoptics: tear, cache, re-solve only the mutant
gaframe · polyphase (→ hyper) · uses solver

em/ — Maxwell's equations in quaternion form
nabla (Hamilton ∇) · maxwell (Treatise form) · biquat $\nabla F = J$
extract: field solves → lumped R, L, C, K · tline from a TEM solve
limits: electrical size $d/\lambda \le 1/10$, otherwise ErrNotQuasiStatic
energy: open-system ledger, Poynting / Tellegen balance
lorentz ⟨F Q†⟩ · duality (hypothetical, off by default)
fdtd (local Yee solve) · topology (∇·B, toroids) · probes
Never runs inside Newton / LU: it supplies parameters and checks

solver/ — linear & nonlinear solve
matrix: dense (gonum LU, < ~30 unknowns) · sparse (Gilbert–Peierls, symbolic reuse)
newton: damping + limiting · reltol 1e-3 · itl1 100 / itl4 10 → ErrNoConvergence
convergence: Newton → gmin stepping → source stepping → pseudo-transient

output/ — results
result: in-memory vectors + energy ledger
csv · raw (SPICE3 binary / ASCII)
one Result type shared by MNA, kron and em

Result → checks

hyper/ — hypercomplex numbers (gonum only)
quat: quaternion phasors and rotors (wraps gonum num/quat)
ga: Clifford algebra Cl(p,q,r), bitmask blades · G2, G3 + sparse multivector
Never imported by solver or device: inner loop stays float64 / complex128

device/ — one file per device family, stamping through StampContext
passive R C L K · source V I (DC PULSE SIN PWL EXP AC) · environment (photovoltaic, thermoelectric) · controlled E F G H · switch S W
arc (Cassie, Mayr, plasma channel → plasma) · diode (SPICE3 level 1) · bjt (Gummel–Poon) · mosfet levels 1–3 (later)
Device: Setup · Load · LoadAC · Converged · Truncate — every device embeds device.Base, so it carries the full external port set

circuit/ — in-memory circuit model
nodes, elements, models · builder.go: programmatic API used by evo
(no text parsing) · Validate() before every simulation

netlist/ — SPICE text input
lexer · parser · .subckt · .param · .tear
units.go: f p n u m k meg g t mil

arc.go

plasma/ (gonum only)
saha · conductivity (Spitzer)
channel energy balance
breakdown · radiation

fluid/ (gonum only)
media: H_2O and D_2O tables
bubble: R–P, Keller–Miksis, Gilmore
acoustic · sonolum

exchange/ — external energy ports on every element (leaf, stdlib only · imported by device, env, em, analysis and evo)
Every element exposes one ExternalPort per registered channel. No model means exactly 0 W, listed as NotModeled. Every model cites a source. In Kron's terms each port is an open path to the exterior junction Ω of its channel.

Electrical	Light	Thermal	Magnetic field	Electric field	Mechanical	Acoustic	Ionizing radiation	Chemical	Neutrino	Gravity / frame	Magnetic charge	Future 1 … N
modeled (MNA)	where defined	modeled	with geometry	with geometry	where defined	where defined	modeled	not modeled yet	computed, ≈ 0	transform, no energy	hypothetical only	reserved

Evaluating one candidate

Genome → circuit.Builder → Validate() → Setup: fix sparsity → Newton + LU → output.Result → ledger + measure → fitness

env inputs and em parameters enter at the Builder · every evaluation runs under a context.Context deadline
any error → typed sentinel (errors.Is) → penalty fitness, so the GA learns from broken circuits

Open-system energy ledger

$dW_{stored}/dt = P_{in} - P_{out} - P_{dissipated}$

Kept per element × channel, at every operating point and timestep
Residual above integration tolerance ⇒ reported as an error
Model-supplied (E/F/G/H) and hypothetical power in their own columns
Surplus is never credited to neutrinos
Efficiency, harvested energy, standby loss → GA fitness
Twin check: Kron open-path power = MNA ledger (1e-9)

Design rules

Never panic, hang or exhaust memory: typed errors only
Deterministic: seeded rand, results sorted before selection
Parallel: no shared mutable state, one simulator per worker
Lumped (quasi-static) limit of Maxwell's equations, checked
Solver core float64 / complex128, allocation-free Load and LU
Golden tests vs. ngspice; twin tests for Kron, GA, quaternions
Pure Go, no cgo: gonum, rwl/lufact, stdlib

Build order: M1 exchange + netlist + R/V/I + dense MNA + .OP → M2 .TRAN → M3 .AC/.DC → M4 diode + Newton → M5 evo → M6 sparse LU → M7 BJT → M8 topology genome → M9 MOSFET, kron, hyper, em, env, plasma, fluid → M10 QSpice cross-check